

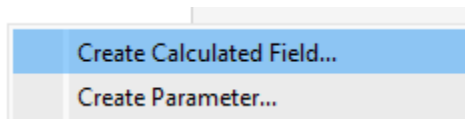
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – IX

Roll No.:T039	Name:Bhavesh Ramakant Salaskar
Class:IT	Batch:1
Date of Assignment:01-09-2026	Date/Time of Submission:01-09-2026

A: Identify and Handle Missing or Inconsistent Data

Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)



Physics_Clean

```
FLOAT(REPLACE(STR([Physics]), '%', ''))
```

Attendance_Clean

```
FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))
```

Step 3: Handle Missing (Null) Numeric Values via Imputation

Math_Clean

```
IFNULL([Math], 0)
```

Chemistry_Clean

```
IFNULL([Chemistry], 0)
```

Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

The screenshot shows a data management interface. On the left, a table list under the heading "Tables" includes entries like "Attendance", "Department", "Full Name", "Student ID", "Measure N.", "Attendance", "Chemistry", "Chemistry_", "Math", "Math_Clear", "Physics", and "Physics_Clean". The "Physics_Clean" entry is highlighted in green. A context menu is open over this entry, listing actions such as "Edit...", "Duplicate", "Rename", "Hide", "Delete", "Create", "Convert to Discrete", "Convert to Dimension", "Change Data Type", "Default Properties", "Geographic Role", "Group by", "Folders", "Replace References...", and "Describe...". The "Edit..." option is currently selected and highlighted in blue. To the right of the menu, a text input field contains the name "Physics_Clean". Below this field, a SQL query is displayed: `IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)`.

B: Convert Wide Data into Tall Format

# Sem5_Scores.csv Physics	# Sem5_Scores.csv Chemistry		
69.0000	95		
90.0000	80		
75.0000	80		
82.0000	86		
88.0000	90	96	95 %

Rename
Copy Values
Hide
Create Calculated Field...
Pivot
Merge Mismatched Fields

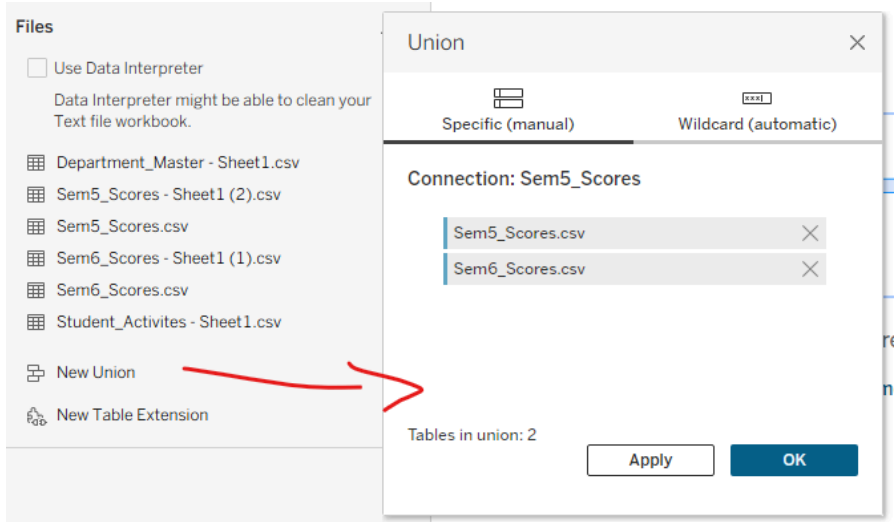
Abc Pivot Pivot Field Names	# Pivot Pivot Field Values
Chemistry	95.0000
Physics	69.0000
Programming	83.0000
Chemistry	80.0000

Abc Pivot Subject	# Pivot Score
Chemistry	95.0000
Physics	69.0000
Programming	83.0000
Chemistry	80.0000
Physics	90.0000

	Clean_Score
# Pivot Score	☒ Number (decimal) ☐ Number (whole)

`IFNULL(FLOAT(REPLACE(STR([Score]), ' ', '')), 0)`

C: Perform Union of Multiple Files



Step 3: Verify the Merged Data Grid

Step 4: Create a Clean Semester Identifier Dimension

Semester

```
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END
```

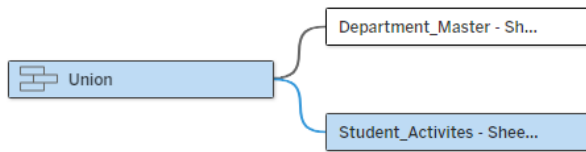
D: Execute Cross-Database Joins



Name Union

Union	Ope...	Departme...
Abc Departi ▼	= ▼	Abc Dept Cc ▼

⊕ Add more fields



Name Student_Activities - Sheet1.csv

Union Ope... Student_A...

Abc Student ▼ = Abc Student ▼

+ Add more fields

Union

Abc Attendance

Abc Department

Abc Full_Name

Abc Semester

Abc Student_ID

Abc Table Name

Attendance_Clean

Chemistry

Chemistry_Clean

Math

Math_Clean

Physics

Physics_Clean

Programming

Union (Count)

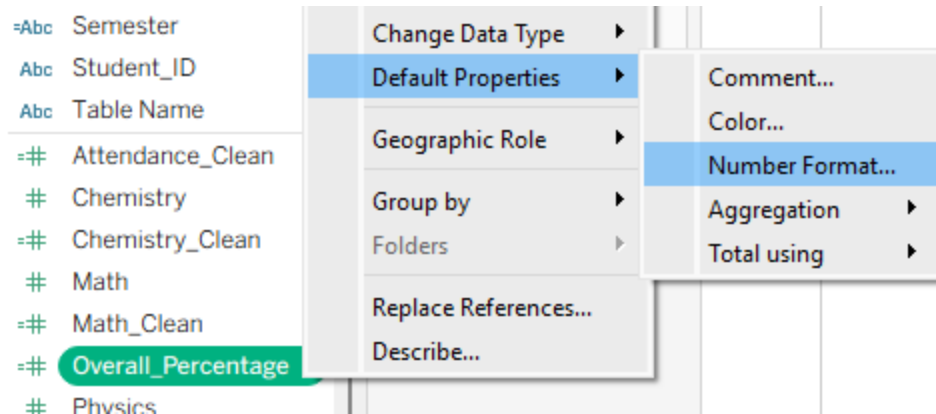
E: Restructure Data for Better Visualization

Total_Score

`[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]`

Overall_Percentage

`[Total_Score] / 400.0`



Default Number Format [Overall_Percentage]

Automatic	Percentage
Number (Standard)	Decimal places:
Number (Custom)	1
Currency (Standard)	
Currency (Custom)	
Scientific	
Percentage	
Custom	

Performance_Band

```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END
```

Attendance_Status

```
IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
ELSE "Shortage Warning (<75%)"
END
```

Pages

Filters

Marks

Automatic

Color

Size

Text

Detail

Tooltip

Performance_...

ATTR(Attenda..

SUM(Overall_...

Columns

Semester

Rows

Student_ID

Full_Name

Sheet 1

Student_ID	Full_Name	Semester
T100	Bhaves	88.8%
T101	Akansha	185.8%
T102	Priyal Verma	86.3%
T103	Rohan Gupta	90.5%
T104	Sneha Patel	91.5%

F: Handle Different Levels of Detail in Datasets

Dept_Average_Percentage_LOD

```
{ FIXED [Department] : AVG([Overall_Percentage]) }
```

Default Number Format [Dept_Average_Percentage_LOD]

Automatic

Number (Standard)

Number (Custom)

Currency (Standard)

Currency (Custom)

Scientific

Percentage

Custom

Percentage

Decimal places:

1

Full_Name

Performance_Band

Semester

Student_ID

Table Name

Attendance_Clean

Chemistry

Chemistry_Clean

Dept_Average_Perc...

Change Data Type

Default Properties

Geographic Role

Group by

Folders

Replace References...

Describe...

T102

Priyal

Comment...

Color...

Number Format...

Aggregation

Total using

T105

Adity

Anan

tudent_Overall_Career_Average

```
{ FIXED [Student_ID] : AVG([Overall_Percentage]) }
```

3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

Variance_From_Dept_Average

```
[Overall_Percentage] - [Dept_Average_Percentage_LOD]
```

3. Set its Number Format to Percentage (1 decimal place).

Benchmark_Status

```
IF [Variance_From_Dept_Average] >= 0 THEN "Above Department  
Benchmark"  
ELSE "Below Department Benchmark"  
END
```

Pages	Columns	Semester
Filters	Rows	Department Student_ID Full_Name
Marks		
Automatic		
Color	Size	Text
Detail	Tooltip	
Benchmark_St..	SUM(Dept_Av..	SUM(Overall_..

Sheet 2			Semester
Department	Student_ID	Full_Name	Other
CS	T107	Rahul Nair	92.0%
			72.5%
ECE	T102	Priyal Verma	86.3%
	T106	Ananya Singh	76.3%
IT	T100	Bhavesb Salaskar	89.0%
	T101	Akansha Ranade	68.8%
			171.5%
			185.8%

G: Apply Advanced Data Cleaning Techniques

Full_Name_Clean

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +  
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -  
1)) +  
" " +  
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1,  
1)) +  
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2,  
LEN(TRIM([Full_Name]))))|
```

First_Name

Last_Name

```
TRIM(SPLIT([Full_Name], " ", 1))| TRIM(SPLIT([Full_Name], " ", -1))
```

Dept_Branch

Section_ID

```
SPLIT([Department], "-", 1) SPLIT([Department], "-", 2)|
```

Student_Display_Label

```
[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"
```

Pages		Columns	Semester
Filters		Rows	Student_Display_Lab..
Dept_Branch: IT		Sheet 3	
Section_ID		Semester	
Marks		Student_Di..	Other
Automatic		88.8%	82.8%
Color Size Text		T100 - Bhavesh Salaskar (IT)	
Detail Tooltip		185.8%	
Performance...		T101 - Akansha Ranade (IT)	
SUM(Overall...		90.5% 64.0%	
		T103 - Rohan Gupta (IT)	

Filter [Dept_Branch]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ I

Enter search text

☐ CS

☐ ECE

☒ IT

☐ MECH